Practical No :5

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Problem statement : Design n-Queens matrix having first Queen placed. Use backtracking to place remaining Queens to generate the final n-queen‘s matrix.

Code :

```
def print_solution(board, n):
    for i in range(n):
        for j in range(n):
            print(board[i][j], end=" ")
        print()


def is_safe(board, row, col, n):
    # Check upper column
    for i in range(row):
        if board[i][col] == 1:
            return False

    # Check upper-left diagonal
    i, j = row - 1, col - 1
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    # Check upper-right diagonal
    i, j = row - 1, col + 1
    while i >= 0 and j < n:
        if board[i][j] == 1:
            return False
        i -= 1
        j += 1

    return True


def solve_n_queens(board, row, n):
    if row == n:
        return True

    for col in range(n):
        if is_safe(board, row, col, n):
            board[row][col] = 1
            if solve_n_queens(board, row + 1, n):
                return True
```

```
            board[row][col] = 0  # Backtrack

    return False


n = int(input("Enter the value of N: "))
board = [[0 for _ in range(n)] for _ in range(n)]

# Place the first queen in the first row
board[0][0] = 1
if solve_n_queens(board, 1, n):
    print("Solution:")
    print_solution(board, n)
else:
    print("No solution exists.")
```

Output :

